AST 6007-01 Spacecraft Dynamics and Control

Final Examination

December 14, 2011

Open Book/Open Notes

Work all problems. In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

1. (15) A quaternion vector given by $\left(\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{2}}{2}\right)$ represents the attitude of a rigid body. Determine the rotation matrix that represents this attitude of the rigid body.

2. (15) A quaternion vector (q_1, q_2, q_3, q_4) satisfies the following rotational kinematic equations

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \frac{1}{2} S(\omega) \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} + \frac{1}{2} q_4 \omega$$

$$\dot{q}_4 = -\frac{1}{2} \omega^T \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}.$$

Show that if the initial values of the quaternion satisfy

$$q_1^2(0) + q_2^2(0) + q_3^2(0) + q_4^2(0) = 1$$

then

$$q_1^2(t) + q_2^2(t) + q_3^2(t) + q_4^2(t) = 1$$

holds for all $t \geq 0$.

3. An inertially symmetric rigid spacecraft has a diagonal inertia matrix given by $I = diag(200, 200, 200) kg - m^2$. It is desired that the spacecraft perform a rotational rest-to-rest maneuver that transfers the spacecraft from its initial attitude given by a rotational matrix representation $R = \exp[S(a)]$, where the unit vector $a = (0.424, -0.566, 0.707)$, to the final attitude given by the rotational matrix R that is the identity. The rotational maneuver should be a rotation about the fixed axis defined by the unit vector a (it is an Euler rotation), and the rotational acceleration about that fixed axis should be either $+0.01\ rad/sec^2$ or $-0.01\ rad/sec^2$. Assume gravity gradient moments can be ignored.

   a. Give an analytical expression for the time dependence of each component of the spacecraft angular velocity vector expressed in the body-fixed coordinate frame.

   b. Give an analytical expression for the time dependence of each component of the control moment vector expressed in the body-fixed coordinate frame.

c. How long is a time period required to complete the rotational maneuver?

4. Consider the pitch axis control model for a spacecraft in a circular orbit given by

$$I_y \Delta\ddot{\theta} = 3\omega_0^2(-I_x + I_z)\Delta\theta + M_y$$

where the orbital velocity vector $\omega_0 = 0.0012\, rad/sec$, the spacecraft principal moments of inertia are $I = diag(150, 200, 100) kg - m^2$. In this equation $\Delta\theta$ denotes the pitch angle and M_y denotes the pitch axis control moment. It is desired to perform a rest-to-rest maneuver using the pitch axis torques that transfers the initial pitch angle $\Delta\theta = 45$degrees to the final pitch angle $\Delta\theta = 0$degrees.

It is claimed that this rest-to-rest maneuver can be performed using a single impulse supplied by the pitch axis control moment.

Is this claim true or false?

If the claim is true, then describe at what time instant the impulse should be applied and describe the required change in the pitch rate due to this impulse.

If the claim is false, justify that it is not possible to accomplish the maneuver using a single impulse.

5. Consider the pitch axis control model for a spacecraft in a circular orbit given by

$$I_y \Delta\ddot{\theta} = 3\omega_0^2(-I_x + I_z)\Delta\theta + M_y$$

where the orbital frequency $\omega_0 = 0.0012\, rad/sec$, the spacecraft principal moments of inertia are $I = diag(150, 100, 200) kg - m^2$. In this equation $\Delta\theta$ denotes the pitch angle and M_y denotes the pitch axis control moment.

a. If the pitch axis control moment $M_y = 0$, show that the pitch equilibrium $(\Delta\theta, \Delta\dot{\theta}) = (0, 0)$ is an unstable equilibrium

b. Show that linear state feedback to the pitch axis control moment can be used to stabilize the pitch axis equilibrium $(\Delta\theta, \Delta\dot{\theta}) = (0, 0)$ of the spacecraft. Choose feedback gains so that the equilibrium of the closed loop is asymptotically stable with damping ratio of 0.7 and natural frequency of $0.008 rad/sec$ (that is, the closed loop characteristic polynomial is $s^2 + 0.0112s + 0.000064$).

AST 6007-01 Spacecraft Dynamics and Control

Final Examination

December 9, 2013 2:10-3:40 PM

Close Books/Close Notes

In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

1. (20) A spacecraft is instantaneously located at the midway point between the Earth and Moon and the spacecraft has zero velocity vector with respect to the Earth-Moon barycentric coordinate frame. Based on the non-dimensionalized equations of motion for the restricted three-body dynamics, answer the following questions:

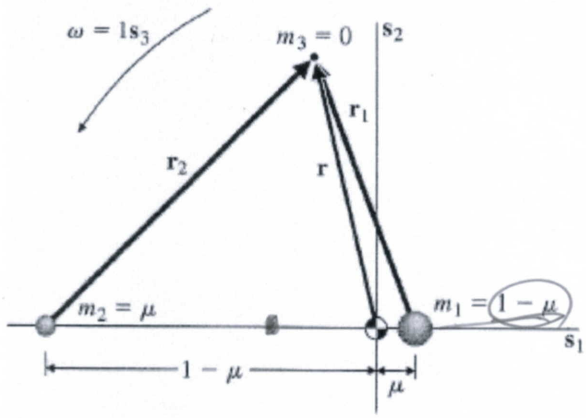

FIGURE 10.1
Geometry of the restricted problem.

$$\ddot{x} - 2\dot{y} - x = -\frac{(1-\mu)(x-\mu)}{r_1^3} - \frac{\mu(x+1-\mu)}{r_2^3}$$

$$\ddot{y} + 2\dot{x} - y = -\frac{(1-\mu)y}{r_1^3} - \frac{\mu y}{r_2^3}$$

$$\ddot{z} = -\frac{(1-\mu)z}{r_1^3} - \frac{\mu z}{r_2^3}$$

$$r_1 = \sqrt{(x-\mu)^2 + y^2 + z^2}, \quad r_2 = \sqrt{(x+1-\mu)^2 + y^2 + z^2}$$

$\mu = 0.0123$ for the Earth-Moon System

$\dot{x} = \dot{y} = \dot{z} = 0$

A. (15) Is the spacecraft in equilibrium at this midway point between the Earth and Moon? If yes, provide your rationale. If not, what is the acceleration vector of the spacecraft with respect to the barycentric coordinate frame?

B. (5) What is the constant (control) acceleration vector that is required to keep the spacecraft in equilibrium at this midway point between the Earth and Moon? Express your control acceleration vector in the barycentric coordinate frame.

2. (20) For the 3-2-1 rotation, derive the angular velocity components $\boxed{\vec{\omega} = [\omega_x \;\; \omega_y \;\; \omega_z]^T}$ in terms of the Euler angles (ψ, θ, ϕ) and $\boxed{\text{their rates.}}$

3. (30) Euler's Equations of Motion

   A. (15) Starting from Newton's law that the inertial time rate of change of the angular momentum is the total applied torque, derive Euler's equations of motion in a body fixed frame, which is assumed to be principal. The moment of inertia in the given set of principal axes, the angular velocity component and the external moment component are given as $I = diag(A,B,C)$, $\vec{M} = [M_1 \;\; M_2 \;\; M_3]^T$ and $\vec{\omega} = [\omega_1 \;\; \omega_2 \;\; \omega_3]^T$, respectively. $I\vec{\omega}(\;\;)$ 3x1

   B. (15) In case the rotational motion is torque-free, show that the kinetic energy and the magnitude of angular momentum are conserved. // 3x3 x 3x1

4. (20) Consider the pitch axis control model for a spacecraft in a circular orbit given by

$$\boxed{I_y \Delta\ddot{\theta} = 3\omega_0^2(-I_x + I_z)\Delta\theta + M_y}$$

where the orbital frequency $\omega_0 = 0.0012\, rad/sec$, the spacecraft principal moments of inertia are $I = diag(150, 100, 200)kg - m^2$. In this equation $\Delta\theta$ denotes the pitch angle and M_y denotes the pitch axis control moment.

   A. If the pitch axis control moment $M_y = 0$, show that the pitch equilibrium $(\Delta\theta, \Delta\dot{\theta}) = (0,0)$ is an unstable equilibrium

   B. Show that linear state feedback to the pitch axis control moment can be used to stabilize the pitch axis equilibrium $(\Delta\theta, \Delta\dot{\theta}) = (0,0)$ of the spacecraft. Choose feedback gains so that the equilibrium of the closed loop is asymptotically stable with damping ratio of 0.7 and natural frequency of $0.008 rad/sec$ (that is, the closed loop characteristic polynomial is $\boxed{s^2 + 0.0112s + 0.000064}$). // $\omega_n =$

5. (10) Briefly discuss rotation matrix (direction cosine matrix), Euler angles and quaternions.

위 3개를 비교해서 사용하는지 //

$I_x \dot{\omega}_x = (I_y - I_z)\omega_y \omega_z + M_x$

$I_y \dot{\omega}_y = (I_z - I_x)\omega_z \omega_x + M_y$

$I_z \dot{\omega}_z = (I_x - I_y)\omega_x \omega_y + M_z$

$$\boxed{\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 0 & & \\ & 0 & \\ & & 0 \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}}$$

즉 $(\Delta\theta, \Delta\dot{\theta}) = (0,0)$ 은 equilibrium 이고,

여기에서의 stability는 $\Delta\theta$를 살린 characteristic e.q 을 봐야 하는 것이다.

즉 $\boxed{S^2 = \dfrac{3\omega_0^2(-I_x + I_z)}{I_y}}$ 가 되는 것이다.

AST 6007-01 Spacecraft Dynamics and Control

송 영범.

Final Examination

December 15, 2014 1:00-2:50 PM

Close Books/Close Notes

In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

$\omega_0 = 2\pi/24$

1. (40) Consider a satellite initially on a geosynchronous circular equatorial orbit whose constant orbital frequency is denoted as $\boxed{\vec{\omega} = \omega\vec{k}}$. After performing a maneuver, it is now located at the following relative position and velocity with respect to its non-maneuvering position in the local-vertical-local-horizontal (LVLH) $\vec{i}\vec{j}\vec{k}$ -frame attached to the geosynchronous orbit:

$$(x, y, z) = (-110, 70, 10)km$$
$$(\dot{x}, \dot{y}, \dot{z}) = (-10, 20, 1)m/\sec$$

If the satellite is to initiate a two-impulse maneuver to return to its original non-maneuvering location in 2 hours, what are the changes in velocities at the initial and terminal time, i.e., $\Delta\vec{v}_0$ and $\Delta\vec{v}_f$?

Hint: The solution to the Hill-Clohessy-Wiltshire (HCW) equation is given as

$$x(t) = \frac{\dot{x}_0}{\omega}\sin(\omega t) - \left(3x_0 + \frac{2\dot{y}_0}{\omega}\right)\cos(\omega t) + \left(4x_0 + \frac{2\dot{y}_0}{\omega}\right)$$
$$y(t) = \left(6x_0 + \frac{4\dot{y}_0}{\omega}\right)\sin(\omega t) + \frac{2\dot{x}_0}{\omega}\cos(\omega t) - (6\omega x_0 + 3\dot{y}_0)t + \left(y_0 - \frac{2\dot{x}_0}{\omega}\right)$$
$$z(t) = \frac{\dot{z}_0}{\omega}\sin(\omega t) + z_0\cos(\omega t)$$

If the relative position in the HCW equation is equal to zero after time t, the initial position and velocity must satisfy the following relation:

$$\dot{y}_0 = \frac{[6x_0(\omega t - \sin(\omega t)) - y_0]\omega\sin(\omega t) - 2\omega x_0(4 - 3\cos(\omega t))(1 - \cos(\omega t))}{(4\sin(\omega t) - 3\omega t)\sin(\omega t) + 4(1 - \cos(\omega t))^2}$$
$$\dot{x}_0 = -\frac{\omega x_0(4 - 3\cos(\omega t)) + 2(1 - \cos(\omega t))\dot{y}_0}{\sin(\omega t)}$$
$$\dot{z}_0 = -z_0\omega COT(\omega t)$$

2. (20) Consider the Euler's rotational equations of motion represented in the principal axes of body-fixed frame:

$$\begin{cases} I_x\dot{\omega}_x = (I_y - I_z)\omega_y\omega_z + M_x \\ I_y\dot{\omega}_y = (I_z - I_x)\omega_x\omega_z + M_y \\ I_z\dot{\omega}_z = (I_x - I_y)\omega_x\omega_y + M_z \end{cases}$$

A. (6) For a general asymmetric spacecraft, show that $(\omega_x \equiv 0, \omega_y \equiv 0, \omega_z \equiv \bar{\omega}_z = \text{constant})$ is an equilibrium solution with no external moments.

B. (8) Linearize the Euler's rotational equations of motion with respect to the equilibrium solution given in (a).

C. (8) Derive the characteristic equation for two non-trivial equations obtained in (b) and their associated eigenvalues case-by-case (i.e., depending on the magnitude of moment of inertia component).

3. (20) Consider the pitch axis control model for a spacecraft in a circular orbit given by

$$I_y \Delta\ddot{\theta} = 3\omega_0^2(-I_x + I_z)\Delta\theta + M_y$$

where the orbital frequency $\omega_0 = 0.0012 rad/\sec$, the spacecraft principal moments of inertia are $I = diag(150, 100, 200) kg - m^2$. In this equation $\Delta\theta$ denotes the pitch angle and M_y denotes the pitch axis control moment.

A. If the pitch axis control moment $M_y = 0$, show that the pitch equilibrium $(\Delta\theta, \Delta\dot{\theta}) = (0,0)$ is an unstable equilibrium

B. Show that linear state feedback to the pitch axis control moment can be used to stabilize the pitch axis equilibrium $(\Delta\theta, \Delta\dot{\theta}) = (0,0)$ of the spacecraft. Choose feedback gains so that the equilibrium of the closed loop is asymptotically stable with damping ratio of 0.7 and natural frequency of $0.008 rad/\sec$ (that is, the closed loop characteristic polynomial is $s^2 + 0.0112s + 0.000064$).

4. (20) A quaternion vector (q_1, q_2, q_3, q_4) satisfies the following rotational kinematic equations

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \frac{1}{2} S(\omega) \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} + \frac{1}{2} q_4 \omega$$

$$\dot{q}_4 = -\frac{1}{2}\omega^T \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}.$$

Show that if the initial values of the quaternion satisfy

$$q_1^2(0) + q_2^2(0) + q_3^2(0) + q_4^2(0) = 1$$

then

$$q_1^2(t) + q_2^2(t) + q_3^2(t) + q_4^2(t) = 1$$

holds for all $t \geq 0$.

이정아

AST 6007-01 Spacecraft Dynamics and Control

Final Examination

December 2, 2015 1:00-3:00 PM

Close Books/Close Notes

In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

1. (20) Suppose that a spacecraft with mass $m=100kg$ operates near a circular orbit with orbital radius $r=6{,}578km$. The Clohessy-Wiltshire equations can describe its orbital perturbation dynamics in the LVLH $\vec{\imath}\,\vec{\jmath}\,\vec{k}$-frame. Under some conditions, the Clohessy-Wiltshire equations can be further approximated by

$$\begin{aligned} m\Delta\ddot{x} &= F_x \\ m\Delta\ddot{y} &= F_y \\ m\Delta\ddot{z} &= F_z \end{aligned}$$

Here $(\Delta x, \Delta y, \Delta z)$ are perturbations of the spacecraft position from the circular orbit, and (F_x, F_y, F_z) are propulsive force components expressed in the LVLH $\vec{\imath}\,\vec{\jmath}\,\vec{k}$-frame. This approximation is often acceptable over a sufficiently small time period $T>0$.

a. (10) Estimate how small the maneuver time period T should be so that the above equations are a satisfactory approximation of the Clohessy-Wiltshire equations within accuracy of about 3%. Hint: Consider the following analytic solutions to the Clohessy-Wiltshire equations:

$$x(t) = \frac{\dot{x}_0}{\omega}\sin(\omega t) - \left(3x_0 + \frac{2\dot{y}_0}{\omega}\right)\cos(\omega t) + \left(4x_0 + \frac{2\dot{y}_0}{\omega}\right)$$

$$y(t) = \left(6x_0 + \frac{4\dot{y}_0}{\omega}\right)\sin(\omega t) + \frac{2\dot{x}_0}{\omega}\cos(\omega t) - (6\omega x_0 + 3\dot{y}_0)t + \left(y_0 - \frac{2\dot{x}_0}{\omega}\right)$$

$$z(t) = \frac{\dot{z}_0}{\omega}\sin(\omega t) + z_0\cos(\omega t)$$

$x(t) - \hat{x}(t) < 0.03 \times x(t)$

$0.97\, x(t) < x(t)$

b. (10) Using the above approximation, a two-impulse rendezvous maneuver is to transfer the initial spacecraft position vector $\Delta\vec{r}(0) = -10\vec{\imath}km$ and initial velocity vector $\Delta\dot{\vec{r}}(0) = 0$ to the terminal spacecraft position vector $\Delta\vec{r}(60sec) = 0$ and terminal velocity vector $\Delta\dot{\vec{r}}(60sec) = 0$. What is the change in spacecraft velocity vector that occurs at each of the impulsive burns?

2. (20) Consider the equations of motion for the Earth-Moon circular restricted three-body problem:

$$\begin{cases} \ddot{x} - 2\omega_0 \dot{y} - \omega_0^2 x = -\dfrac{\mu_E (x - D_E)}{r_E^3} - \dfrac{\mu_M (x + D_M)}{r_M^3} + \dfrac{F_x}{m} \\ \ddot{y} + 2\omega_0 \dot{x} - \omega_0^2 y = -\dfrac{\mu_E y}{r_E^3} - \dfrac{\mu_M y}{r_M^3} + \dfrac{F_y}{m} \\ \ddot{z} = -\dfrac{\mu_E z}{r_E^3} - \dfrac{\mu_M z}{r_M^3} + \dfrac{F_z}{m} \end{cases}, \quad \begin{cases} r_E^2 = (x - D_E)^2 + y^2 + z^2 \\ r_M^2 = (x + D_M)^2 + y^2 + z^2 \\ \mu_E = 398{,}601 km^3/\sec^2 \\ \mu_M = 4{,}887 km^3/\sec^2 \\ D_E = 4{,}674 km \\ D_M = 380{,}073 km \\ \omega_0 = 2.66199 \times 10^{-6} rad/\sec \end{cases}$$

a. (10) Linearize the above equations of motion with respect to the Lagrangian point $L_2 : (\bar{x} = -444{,}645.6 km, \bar{y} = 0, \bar{z} = 0)$.

b. (5) What are the eigenvalues of the linearized out-of-plane equations of motion?

c. (5) Discuss the stability of out-of-plane motion based on its eigenvalues.

3. (30) Euler's Equations of Motion

a. (15) Starting from Newton's law that the inertial time rate of change of the angular momentum is the total applied torque, derive Euler's equations of motion in a body fixed frame, which is assumed to be principal. The moment of inertia in the given set of principal axes, the angular velocity component and the external moment component are given as $I = diag(A, B, C)$, $\vec{M} = [M_1 \;\; M_2 \;\; M_3]^T$ and $\vec{\omega} = [\omega_1 \;\; \omega_2 \;\; \omega_3]^T$, respectively.

b. (15) In case the rotational motion is torque-free, show that the kinetic energy and the magnitude of angular momentum are conserved.

4. (30) Consider the pitch axis control model for a spacecraft in a circular orbit given by

$$I_y \Delta\ddot{\theta} = 3\omega_0^2(-I_x + I_z)\Delta\theta + M_y$$

where the orbital frequency $\omega_0 = 0.0012\, rad/sec$, the spacecraft principal moments of inertia are $I = diag(150, 100, 200) kg - m^2$. In this equation, $\Delta\theta$ denotes the pitch angle and M_y denotes the pitch axis control moment.

a. (15) If the pitch axis control moment $M_y = 0$, show that the pitch equilibrium $(\Delta\theta, \Delta\dot{\theta}) = (0, 0)$ is an unstable equilibrium

b. (15) Show that linear state feedback to the pitch axis control moment can be used to stabilize the pitch axis equilibrium $(\Delta\theta, \Delta\dot{\theta}) = (0, 0)$ of the spacecraft. Choose feedback gains so that the equilibrium of the closed loop is asymptotically stable with damping ratio of *0.7* and natural frequency of $0.008 rad/sec$ (that is, the closed loop characteristic polynomial is $s^2 + 0.0112s + 0.000064$).

AST 6007-01 Spacecraft Dynamics and Control

Final Examination

December 16, 2016

*Close Books/Notes: 3:05-3:55 PM

In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

1. (30) Consider the motion of a spacecraft subject to the gravity of the Earth modelled as a point mass and its own thrust. Let m be the mass of the spacecraft, $\mu = GM$ be the gravitational parameter of the Earth where G is Newton's gravitational constant and M is the mass of the Earth, $\vec{r}$ be the position vector of the spacecraft, and $\vec{F}$ be the thrust.

   A. (5) Let $\vec{r} = xi + yj + zk$, $\vec{F} = F_x i + F_y j + F_z k$ and the state vector $\vec{x} = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \end{bmatrix}^T$ $= \begin{bmatrix} x & \dot{x} & y & \dot{y} & z & \dot{z} \end{bmatrix}^T$ be defined in the Cartesian coordinate system. Describe the equations of motion in the Cartesian coordinate system in state space form.

   B. (10) Let $\vec{r} = r\vec{e}_r$ and $\vec{F} = F_r\vec{e}_r + F_v\vec{e}_v + F_\lambda\vec{e}_\lambda$ be defined in the spherical coordinate system with their components in the order of radial, longitudinal, and latitude direction, and $\vec{\omega}$ be the angular velocity vector of the spacecraft. Describe the equations of motion in the spherical coordinate system.

   C. (15) Linearize the radial (first) equations of (B) with respect to the following operation point:

$$\begin{cases} r \equiv R > 0 \\ v = \sqrt{\dfrac{\mu}{R^3}}t + C, \ C = \text{constant} \\ \lambda \equiv 0 \end{cases}$$

$r = R + \Delta r$

$v = \sqrt{\frac{\mu}{R^3}}\, t + \Delta v$

$\lambda = 0 + \Delta\lambda$

2. (20) Consider the Euler's rotational equations of motion represented in the principal axes of body-fixed frame:

$$\begin{cases} I_x\dot{\omega}_x = (I_y - I_z)\omega_y\omega_z + M_x \\ I_y\dot{\omega}_y = (I_z - I_x)\omega_x\omega_z + M_y \\ I_z\dot{\omega}_z = (I_x - I_y)\omega_x\omega_y + M_z \end{cases}$$

   A. (5) For a general asymmetric spacecraft, show that $(\omega_x \equiv 0, \omega_y \equiv 0, \omega_z \equiv \bar{\omega}_z = \text{constant})$ is an equilibrium solution with no external moments.

   B. (5) Linearize the Euler's rotational equations of motion with respect to the equilibrium solution given in (A).

   C. (10) Derive the characteristic equation for two non-trivial equations obtained in (B) and their associated eigenvalues case-by-case (i.e., depending on the magnitude of moment of inertia component).

5

*Open Books/Notes: 4:00-5:55 PM

3. (25) An inertially symmetric rigid spacecraft has a diagonal inertia matrix given by $I = diag(200, 200, 200) kg - m^2$. It is desired that the spacecraft perform a rotational rest-to-rest maneuver that transfers the spacecraft from its initial attitude given by a rotational matrix representation $R = \exp[S(a)]$, where the unit vector $a = (0.424, -0.566, 0.707)$, to the final attitude given by the rotational matrix R that is the identity. The rotational maneuver should be a rotation about the fixed axis defined by the unit vector a (it is an Euler rotation), and the rotational acceleration about that fixed axis should be either $+0.01\, rad/sec^2$ or $-0.01\, rad/sec^2$. Assume gravity gradient moments can be ignored.

   A. Give an analytical expression for the time dependence of each component of the spacecraft angular velocity vector expressed in the body-fixed coordinate frame.

   B. Give an analytical expression for the time dependence of each component of the control moment vector expressed in the body-fixed coordinate frame.

   C. How long is a time period required to complete the rotational maneuver?

4. (25) Consider the pitch axis control model for a spacecraft in a circular orbit given by

$$I_y \Delta\ddot{\theta} = 3\omega_0^2(-I_x + I_z)\Delta\theta + M_y$$

where the orbital velocity vector $\omega_0 = 0.0012\, rad/sec$, the spacecraft principal moments of inertia are $I = diag(150, 200, 100) kg - m^2$. In this equation $\Delta\theta$ denotes the pitch angle and M_y denotes the pitch axis control moment. It is desired to perform a rest-to-rest maneuver using the pitch axis torques that transfers the initial pitch angle $\Delta\theta = 45$degrees to the final pitch angle $\Delta\theta = 0$degrees.

It is claimed that this rest-to-rest maneuver can be performed using a single impulse supplied by the pitch axis control moment.

Is this claim true or false?

If the claim is true, then describe at what time instant the impulse should be applied and describe the required change in the pitch rate due to this impulse.

If the claim is false, justify that it is not possible to accomplish the maneuver using a single impulse.

AST 6007-01 Spacecraft Dynamics and Control

Final Examination

December 11, 2017 3:00-5:00 PM

Close Books/Close Notes

In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

1. (40) The equations of motion for the Moon-Earth circular restricted three-body problem is given as follows:

$$\begin{cases} \ddot{x} - 2\omega_0 \dot{y} - \omega_0^2 x = -\dfrac{\mu_E (x - D_E)}{r_E^3} - \dfrac{\mu_M (x + D_M)}{r_M^3} + \dfrac{F_x}{m} \\ \ddot{y} + 2\omega_0 \dot{x} - \omega_0^2 y = -\dfrac{\mu_E y}{r_E^3} - \dfrac{\mu_M y}{r_M^3} + \dfrac{F_y}{m} \\ \ddot{z} \qquad\qquad\qquad = -\dfrac{\mu_E z}{r_E^3} - \dfrac{\mu_M z}{r_M^3} + \dfrac{F_z}{m} \end{cases}, \quad \begin{cases} r_E^2 = (x - D_E)^2 + y^2 + z^2 \\ r_M^2 = (x + D_M)^2 + y^2 + z^2 \\ \mu_E = 398{,}601 km^3/\sec^2 \\ \mu_M = 4{,}887 km^3/\sec^2 \\ D_E = 4{,}674 km \\ D_M = 380{,}073 km \\ \omega_0 = 2.66199 \times 10^{-6} rad/\sec \end{cases}$$

A. (10) Show that $(x_e \equiv -444{,}645.6, y_e \equiv 0, z_e \equiv 0)$ is an equilibrium solution with no propulsive forces.

B. (20) Linearize the equations of motion with respect to the equilibrium solution given in (A).

C. (10) Discuss the stability of the equilibrium solution given in (A) based on the linearized equation derived in (B).

$J\dot{\vec{\omega}} + \vec{\omega} \times (J\vec{\omega}) = \vec{M}$

$\dot{R} = S(\vec{\omega}) R$

$R = e^{S(\vec{\omega})\phi}$

2. (30) Consider the following Euler's rotational equations of motion and quaternion kinematic equations:

$\vec{H} = J\vec{\omega}$

$$J\dot{\omega} = -S(\omega) J\omega + M$$

$$\dot{\hat{q}} = \frac{1}{2} S(\omega)\hat{q} + \frac{1}{2} q_4 \omega, \quad \dot{q}_4 = -\frac{1}{2}\omega^T \hat{q}$$

$$\omega = [\omega_x \; \omega_y \; \omega_z]^T, \; M = [M_x \; M_y \; M_z]^T, \; q = [q_1 \; q_2 \; q_3 \; q_4]^T = [\hat{q} \; q_4]^T, \; S(\omega) = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

$\vec{M}$을 정해주렴!

A. (10) Determine the nonlinear control law to stabilize the equilibrium ($\omega = [0 \; 0 \; 0]^T$, $q = [0 \; 0 \; 0 \; 1]^T$) based on Lyapunov approach.

B. (10) Show that if the initial values of the quaternion satisfy $\boxed{q_1^2(0) + q_2^2(0) + q_3^2(0) + q_4^2(0) = 1}$ then $q_1^2(t) + q_2^2(t) + q_3^2(t) + q_4^2(t) = 1$ holds for all $t \geq 0$.

C. (10) In case that the rotational motion is torque-free and $J = diag(A, B, C)$, show that the kinetic energy and the magnitude of angular momentum are conserved.

$T = \frac{1}{2}\omega^T J \omega$
$= \frac{1}{2}(J_x \omega_x^2 + \cdots)$

3. (15) The rotation matrix R can be expressed by the quaternion:

$$R(q) = (q_4^2 - \hat{q}^T\hat{q})I_{3\times3} + 2\hat{q}\hat{q}^T - 2q_4 S(\hat{q})$$

When the orientation is slightly changed as the small angle $\Delta\phi$ about the axis $\mathbf{a} = [a_1\ a_2\ a_3]^T$ from the reference frame, derive the rotation matrix R of the infinitesimal rotation as a function of $\Delta\phi$ and $\mathbf{a}$.

Hint: $\sin\phi = 2\sin\frac{\phi}{2}\cos\frac{\phi}{2}$, $\cos\phi = \cos^2\frac{\phi}{2} - \sin^2\frac{\phi}{2}$

시변?

4. (15) Does only one quaternion exist for representing one orientation? Justify your answer.

$\hat{q} = [q_1 q_2 q_3]$

$(q_1 = a_1 \sin\frac{\phi}{2}$

$\vdots)$

$(q_4 = \cos\frac{\phi}{2}\ ?)$

$R = e^{S(\vec{a})\cdot\phi}$

$\dot{R} = S(\vec{\omega})\cdot R$